

Question 1 Consider the functions $f : [0, 1] \rightarrow [a, b]$ and $g : [0, 1] \rightarrow [a, b]$ given by $f(x) = x(b-a) + a$ and $g(x) = \frac{x-a}{b-a}$.

$$g(f(x)) = \frac{x(b-a)+a-a}{b-a} = \frac{x(b-a)}{b-a} = x.$$

Thus, f is invertible, and therefore is a bijection.

Question 2 (a) Consider the function $f : \mathbb{Z}^+ \rightarrow \mathbb{Z} \setminus \{0\}$ given by

$$f(n) = \begin{cases} \frac{n}{2} & n \text{ is even} \\ \frac{-1-n}{2} & \text{otherwise} \end{cases}.$$

(b) *Proof.* We show that g is a bijection by constructing an inverse. Consider the function $m : \mathbb{Q}^+ \rightarrow \mathbb{Z}^+$ defined by $m(l) = p_1^{f^{-1}(a_1)} \cdots p_r^{f^{-1}(a_r)}$. We know f^{-1} exists from (a). Composing m and g , we compute

$$m(g(n)) = p_1^{f^{-1}(f(a_1))} \cdots p_r^{f^{-1}(f(a_r))} = p_1^{a_1} \cdots p_r^{a_r} = n.$$

Thus, g is bijective. $\square$

Question 3 *Proof.* Suppose A, B, C, D are sets, and $A \prec C$. For the sake of contradiction, assume that $D \preceq B$. There are two cases:

($D \approx B$) If $D \approx B$, then by the transitivity of $\approx$, $D \approx A$. However, if this is the case, then we have $D \prec C$ by our assumption that $A \prec C$. This is a contradiction with our initial assumption that $C \approx D$. Thus, $D \not\approx B$.

($D \prec B$) If $D \prec B$, then there exists an injection $D \hookrightarrow B$. By our assumption that $A \approx B$, there is an injection $B \hookrightarrow A$. The composition of injective functions is injective, and thus there is an injection $D \hookrightarrow A$. Thus, $D \preceq A$. If $D \approx A$, we reach a contradiction from case 1. If $D \prec A$, we have $D \prec A \prec C$, which would contradict our initial assumption that $D \approx C$.

We reach a contradiction in both cases, and so we must have that $B \prec D$. $\square$

Question 4 *Proof.* Suppose a set A is infinite, and that $B \subseteq A$ is finite. Assume for the sake of contradiction that $A \setminus B$ is finite. We note that $A \setminus B \cup B = A$. By Theorem ? from the notes, a finite union of finite sets is finite. Thus, $A \setminus B \cup B = A$ is finite—however, this is a contradiction with our assumption that A is infinite. Thus, $A \setminus B$ is infinite. $\square$

Question 5 *Proof.* Suppose a set A is countable, and that $B \subseteq A$ is countable. Assume for the sake of contradiction that $A \setminus B$ is countable. We note that $(A \setminus B) \cup B = A$. By Theorem 7.6 from the notes we have that the union of two countable sets is itself countable. Thus, we must have that A is countable, which is a contradiction to our assumption that A was uncountable. Thus, $A \setminus B$ is uncountable. $\square$